airiti

Journal of Aeronautics, Astronautics and Aviation, Vol. 56, No. 6S, pp. 1081 – 1097 (2024)
DOI: https://doi.org/10.6125/JoAAA.202412_56(6S).06

Numerical Analysis for Aerodynamic Characteristics of the Unmanned Aerial Vehicle (UAV) in Forward Flight

Thi-Hong-Hieu LE [1, 2 *], Khanh NGUYEN [3], Thi-Hong-Nhi VUONG [1, 2]

[1] *Department of Aerospace Engineering, Faculty of Transportation Engineering, Ho Chi Minh City University of Technology (HCMUT), 268 Ly Thuong Kiet Street, District 10, Ho Chi Minh City, Vietnam.*
[2] *Vietnam National University Ho Chi Minh City, Linh Trung Ward, Thu Duc District, Ho Chi Minh City, Vietnam.*
[3] *Department of Smart Vehicle Engineering, Future Drone Center, Konkuk University, Seoul 05029, South Korea.*

ABSTRACT

Cruising with an appropriate wing configuration and propulsion system to maximize aerodynamic performance, flying vehicles can conserve energy and extend their flight duration. This paper presents a series of simulations using OpenFOAM to investigate aerodynamic characteristics of the 3-dimensional (3D) fixed wing, constructed from the BE12355D airfoil, in a novel concept of hybrid unmanned aerial vehicle (UAV) during forward flight. In the design, the aircraft is equipped with a fixed wing for cruising and can achieve vertical takeoff and landing using three rotors mounted on the tri-copter arms. During missions, it is configured to generate 60% of required lift from three rotors, while the remaining 40% is produced by the fixed wing. Using the K-ω Shear Stress Transport (SST) turbulence model, characteristics of the low Reynold number flow are examined without accounting for three rotors. The lift and drag coefficients of the UAV cruising at a speed of 15 m/s are estimated at various angles of attack (AoA). The findings indicate that at AoA of 17°, the predicted maximum lift coefficient of the full 3D UAV is about 1.4, smaller than the 1.5 observed for the 2D airfoil at 10°. Additionally, the lift coefficient curve demonstrates a steeper slope for the 2D airfoil in contrast to the shallower slope of the 3D wing. The velocity vector field illustrates the complicated phenomena of the laminar separation bubbles that characterize the transitional flow over the wingspan, especially in the inboard wing region. This study offers valuable insights for the future investigation of stability and control analyses in flying vehicles, particularly through numerical method.

Keywords: Hybrid UAV, Fixed-wing, Cruising, Aerodynamic performance, Low Reynolds flow, Computational fluids dynamics (CFD), Simulation

I. INTRODUCTION

Unmanned aerial vehicles (UAV) are known as the potential aerial devices for a wide range of applications. Every day, millions of drones are efficiently used to transport packages, work in search and rescue, inspect pipelines or manage traffic systems. As a result, there are so many researchers concerned with controlling and predicting the aerodynamic properties of drones in different flight missions. In order to build a new design of small UAVs, due to the challenges of low Reynold number flow, an integrated process for design and testing is envisaged.

For the first stage of UAV design, the numerical design tools – such as iFly [1], VLM [2] or traditional design methods, can be employed to predict basic stability characteristics. Additionally, the second phase of the aerodynamic flow predictions are conducted through a series of computational fluid dynamics (CFD) simulations

Manuscript received, August 13, 2024, final revision, November 1, 2024
* To whom correspondence should be addressed, E-mail: honghieu.le@hcmut.edu.vn

airiti

using various software such as MEGAFLOW [3], OpenFOAM, and ANSYS Fluent.

In addition to wind tunnel testing, the numerical approach is often employed to analyze the aerodynamic properties of mission-based UAVs operating at low altitudes and low flight speeds. Sun et al. [4] carried out the CFD simulations of a high-curvature wing, S1210, to analyze aerodynamic performance at stalling angle of attack of 18°. Similarly, as reported by Johari et al. [5], the aerodynamic lift, drag, and moment coefficients of the unconventional Hawkeye UAV aircraft with a forward-swept wing configuration were investigated through a series of CFD simulations using ANSYS Fluent. The results were further used to estimate UAV performance and enhance aerodynamic design. Additionally, the CFD method is effectively employed to investigate characteristics of the low Reynolds number flow. According to our previous publication [6], a finite, symmetric teardrop model with the thickness of 40% of the chord was numerically examined at the angle of attack (AoA or α) of 0° in the critical Reynolds regime, where the drag crisis phenomenon occurred at Reynolds numbers (Re) ranging from 4×10^4 to 8×10^4. Furthermore, the study revealed that using the K-ω Shear Stress Transport (SST) transition flow model, the CFD simulations can also detect the occurrence of laminar separation bubbles (LSB) in the critical Reynolds region, which explains a reduction in drag. Specifically, at Re = 5×10^4, the phenomenon of LSB was detected in the inboard wing region, while its presence became less clear toward the tip. In agreement with our previous numerical report, within the range of critical Reynolds number, the nonlinear aerodynamic characteristics and unsteady behaviors of LSB formation, as well as its progressive spanwise variation have been experimentally and numerically confirmed by Schmitz [7], Xia et al. [8], and Toppings and Yarusevych [9].

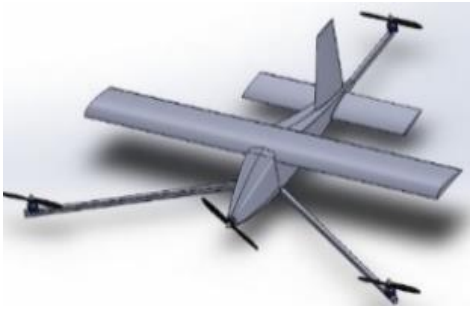


Figure 1 Three-dimensional drawing of UAV-HOPE design

For example, as reported by Xia et al. [8], numerous studies using the CFD method investigated the nonlinear effect and static hysteresis effect of the transition flow regime around 2D airfoils by visualizing the structures of LSB across different Reynolds numbers ranging from 10^4 to 5×10^5. Further, findings from Schmitz [7] indicate that unsteady formation of the LSB structures on the airfoil were observed within the critical Reynolds number range of 10^4 to 10^6. It is noteworthy that the current Reynolds number for our studies falls within the same category as this range. Although the LSB was not consistently observed along the entire wingspan of the 3D tear drop model by CFD simulations, the CFD-based drag coefficient curve aligned with the experimental data reported by Dong et al. [10], which also documented the drag reduction for the same 3D tear drop model within the same range of Reynolds numbers.

Table 1 UAV provisional parameters

Properties	Parameters
Flight altitude	50-100 m
Operating perimeter	3.14 km
Cruising speed – maximum speed	15 m/s – 24 m/s
Reynolds number	~ 142,000
Wingspan, chord, and aspect ratio	b = 1.2 m, c = 0.15 m, and AR = 8
Take-off weight*	3.5 kg (for the total gross weight)
Lift generation by the fixed wing (without rotors)	1.4 kg (as 40% of the total lift)

For the same purpose, this paper presents the CFD method using OpenFOAM to study the aerodynamic characteristics of the viscous flow around the UAV–HOPE design (see Figure 1) during forward flight, excluding the effects of the three rotors. In the design of hybrid UAV [14], the UAV is equipped with the fixed wing for cruising and can achieve vertical takeoff and landing using the three rotors mounted on the tri-copter arms. The lift is generated 60% by the tri-copter and 40% by the fixed wing during cruising, the numerical analysis for the UAV-HOPE project comprises two phases. As reported in [12], we first used OpenFOAM software to numerically evaluate the influence of three rotors on the tricopter's airframe in forward flights. The virtual blade method from the standard library of OpenFOAM was employed to simulate the rotation of the rotor disc. Then, the aerodynamic investigation of UAV-HOPE design is conducted without three rotors' operation. Using the airfoil typed BE12355D to construct the fixed wing, a series of 2D and 3D CFD simulations were conducted to characterize the aerodynamics of the 2D airfoil and 3D wing cruising at a speed of 15 m/s. These simulations were performed using ANSYS and OpenFOAM, as referenced in [13] and [14], respectively. The findings could provide some references for the grid generation in the 3D computational domain and the choice of turbulence models for the current simulations for the full UAV model. Due to lack of reliable validation and sufficient grid size in the previous works in

airiti

[13] and [14], the flow field of the HOPE UAV (without rotors) is simulated by Semi-Implicit Method for Pressure-Linked Equations – (SIMPLE) algorithm. The standard simpleFoam solver in OpenFOAM is employed in this study. The numerical simulation results are analyzed and compared to assess the preliminary design parameters. Furthermore, the detailed visualization of the flow field is also investigated, providing a comprehensive understanding of the flow behavior over the UAV.

The paper is structured as follows: Section 0 introduces the geometry and mesh generation, as well as the definition of the desired boundary-layer thickness. Section 0 covers the boundary conditions, selection of turbulence model, and definition of flow direction for various wing angles of attack. In Section 0, we present the simulation results compared with our previously obtained research. Finally, Section 0 concludes the paper.

II. GEOMETRY AND MESH CONFIGURATION

2.1 Geometry generation

The geometry is firstly created using Salome software, as indicated in Figure 2. The wingspan, b = 1.2 m, is used as the reference length to create the computational domain. In forward flight, the domain is typically modeled as a rectangular shape, measuring 7.2 m in width, 7.2 m in height, and 12.0 m in length, which corresponds to (6×6×12)b, respectively. The distance from the UAV to the outlet boundary is set at 8.5b to ensure that the flow stabilizes as it passes through the UAV. Meanwhile, the distance from the UAV to the inlet boundary is relatively shorter, approximately 1.5b.

As mentioned in [13] and [14], the studies of aerodynamic properties for the 2D airfoil and the 3D fixed wing were acquired with some valuable results. This section summarizes the mesh configurations for the 2D airfoil and 3D wing simulations (see Figures 3 and 4, respectively). These works are considered as references for geometry and mesh generation in the current study. Furthermore, the details of comparison of the lift and drag coefficients by the 2D airfoil, 3D fixed wing, and full UAV model will be discussed in Section 4.2.

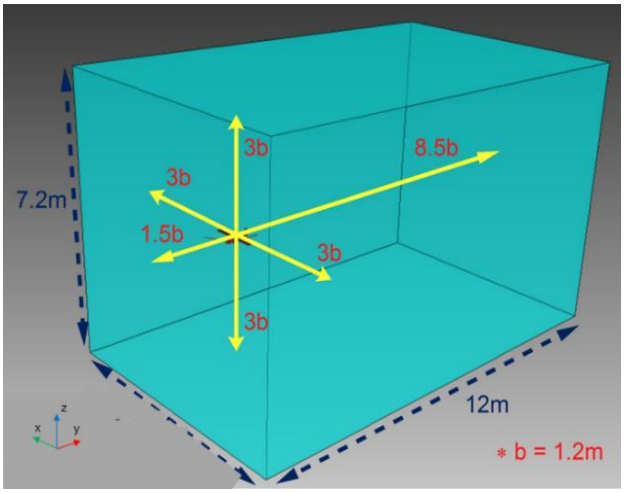


Figure 2 Geometry creation for the simulation of 3D full UAV model

2.2 Mesh generation

This section details the mesh generation process for the full 3D UAV model, as described in Figure 2. This step includes two meshing phases: using Salome software to create the base mesh for the computational domain and employing the snappyHexMesh (sHM) tool for generating the hexahedral mesh of the complex geometry of UAV.

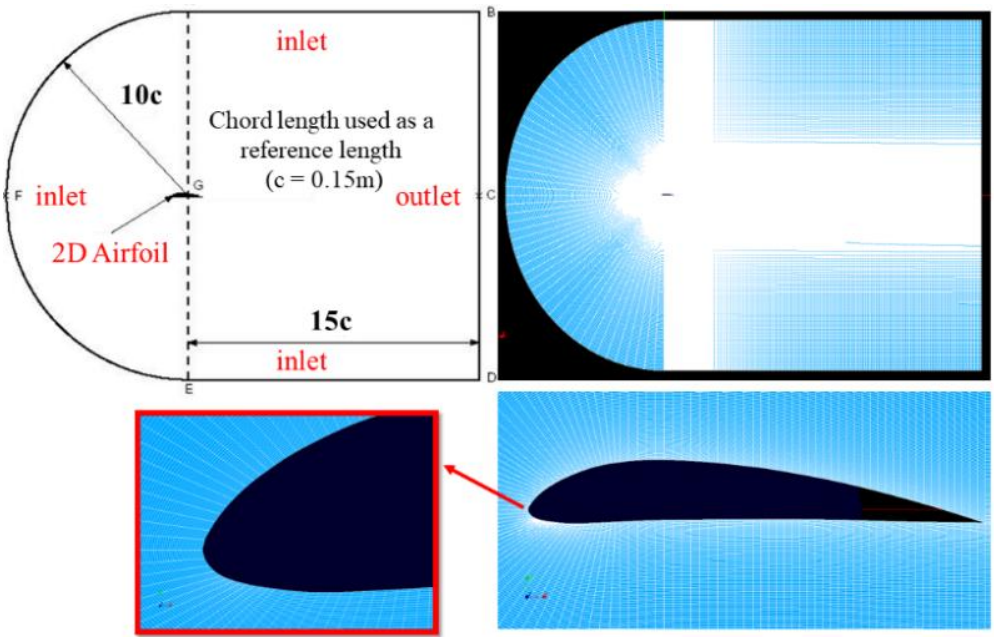


Figure 3 Geometry and mesh generation for the 2D airfoil simulation [13]

airiti

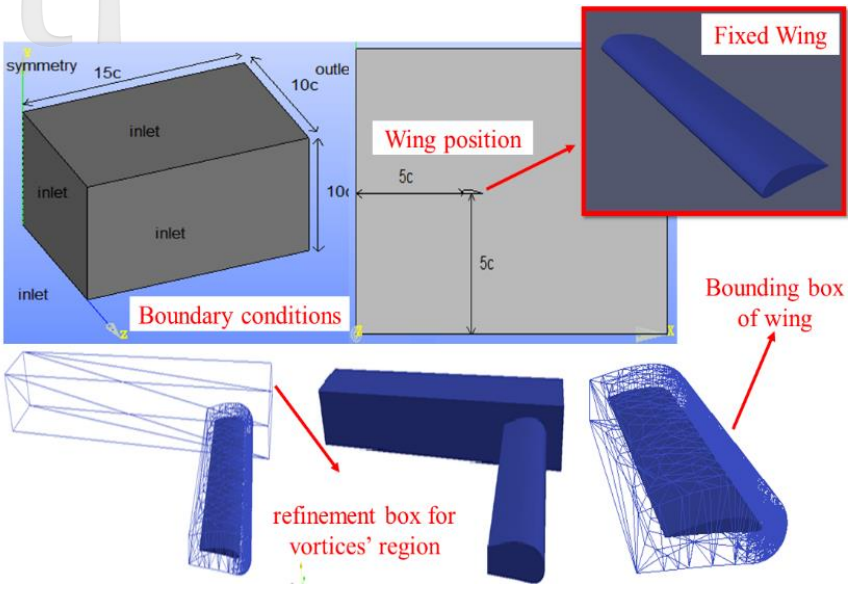


Figure 4 Geometry generation for the fixed wing simulation [14]

2.2.1 The background mesh

The initial preparation of the hexahedral background mesh, conducted using Salome, is essential for achieving the highest quality. To achieve a high-quality mesh, specific criteria must be met, including an aspect ratio of 1, minimal maximum skewness approaching 0, and a non-orthogonality index below 30. At this stage, the base mesh comprises 200,000 hexahedral cell elements.

2.2.2 The refining mesh

In this second stage, the sHM tool, a very useful hexa-mesh generator for complicated geometries, is used to create a structured grid of the full 3D UAV model. The mesh refinement around the UAV is considered by adding the virtual boxes to create the regions with high mesh density. More details on the mesh generation using the sHM tool can be found in [15]. The workflow is summarized as follows: the background mesh is initially refined at the 2D level to capture the geometry features and surfaces, including points and edges extracted from the 3D CAD model files. This is regarded as the cell splitting at the feature edges and surface. After removing the cells inside the wall boundaries, cell splitting is performed in the specified regions by defining the mesh refinement levels. For example, by adding the virtual regions adjacent to the UAV's wall surface, the mesh configuration, consisting of six refinement levels, is shown inFigure 5.

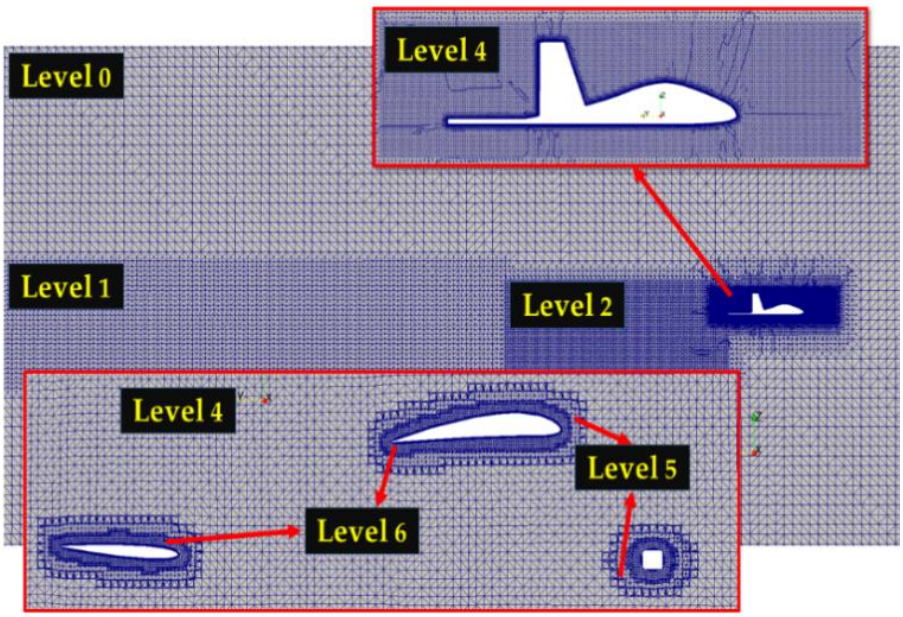


Figure 5 Mesh configuration around the body, tricopter's arm, fixed wing, and horizontal stabilizer shown with varying refining levels (side views). Note that the figures in the red boxes depict the 2D mesh configuration near the UAV at the longitudinal plane of body symmetry (above) and at the location with the horizontal stabilizer, fixed wings, and tricopter's arm (below).

airiti

Table 2 Mesh details

Properties	Fined mesh	Coarse mesh
Number of cells	13.2 million	7.6 million
Aspect ratio	~ 101	~ 32
Average mesh non-orthogonality	6.73	7.73
Skewness	3.8	3.46
Percentage of hexahedral grid elements	96%	94%
First cell thickness	2.5×10^{-5} m	8×10^{-5} m
Number of layers	~ 9.33	~ 7
Total layer thickness	1.67 mm ~ 90% desired thickness	1.47 mm ~ 80% desired thickness

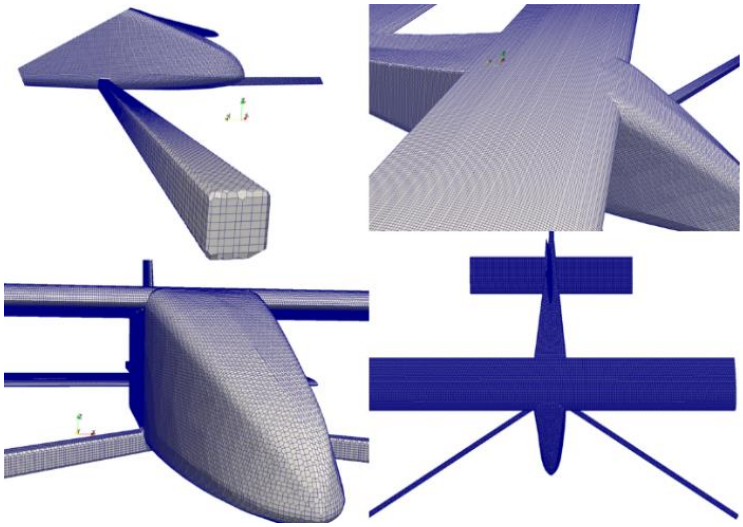


Figure 6 Mesh configuration on the UAV's surface

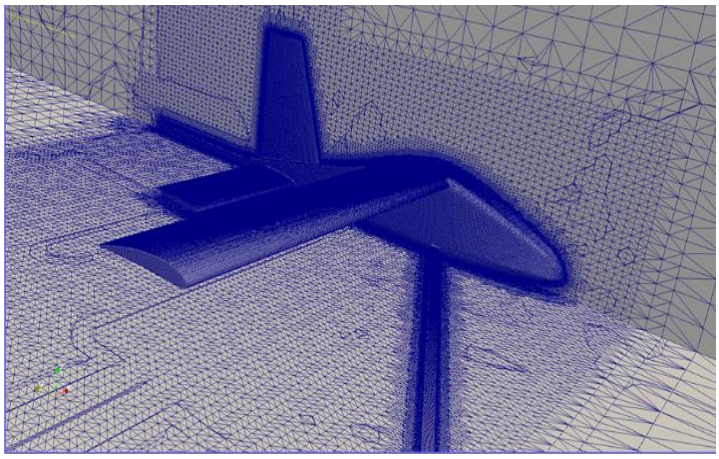


Figure 7 Grid elements in the refining box. The first bounding box is set to the maximum refinement level of 4 (demonstrated inFigure 5)

There are three separated boxes with increasing refinement levels (namely, levels 1, 2 and 4) compared to the zero level of the base mesh, corresponding to the regions extending from the far fields to those surrounding the UAV. Meanwhile, the highest level (level 6) encompasses the wall surface of the UAV. The mesh transition zones between the defined refinement levels are generated by the sHM tool. Finally, the cell vertex points are snapped to the surface of the UAV walls for the output mesh, followed by the optional stage of adding the hexahedral mesh layers aligned to the wall surface.

2.2.3 Grid convergence study

In the grid convergence study, the coarse mesh (7.6 million cells) is created to compare to results from the fined mesh (13 million grid elements). Using the sHM tool, more than 90% of the total grid elements are composed of hexahedral grid cells. The mesh information is summarized in Table 2.

airiti

2.2.4 Boundary layers

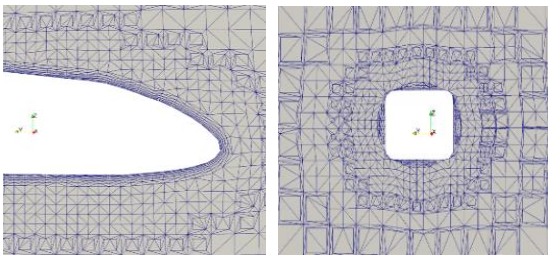
As an optional step performed by the sHM tool to increase the accuracy level, adding the thin boundary layers around the wall surface is considered in this section. As defined, the first cell height is the relative distance of the first cell to the wall surface. If the first layer thickness is significantly small, the quality of the mesh, including the skewness and aspect ratio, becomes very poor. Otherwise, the first cell thickness is set larger, the wall is not properly resolved for low Reynolds number flows. Particularly, a non-dimensional parameter that characterizes the distance from the wall in relation to the viscous sublayer, Y^+, is defined as follows:

$$Y^+ = u \times y \times \nu^{-1}, \quad (1)$$

where u is the friction velocity at the nearest wall (m/s), y represents the distance from the first cell to the wall (m), and ν is the local kinematic viscosity of the fluid (m^2/s). Note that Y^+ value can be reported once the simulation is complete. To calculate the first boundary-layer thickness, a desired Y^+ value is selected. For example, if $Y^+ = 1$, the first layer thickness is approximately 2.5×10^{-5} m using Eq. (1). This indicates that the first layer thickness is significantly small, which leads to poor mesh quality, including issues such as illegal face detections and cells with aspect ratio larger than 100. Moreover, convergence is achieved slowly, when the aspect ratio is large [16].

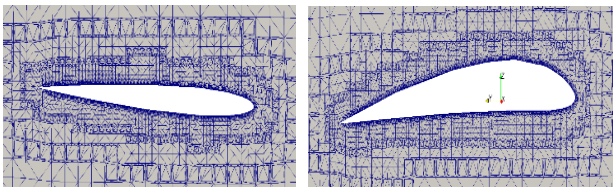
To resolve this problem, the y plus value is increased slightly to reduce the cell average aspect ratio with the same size of the base mesh. A higher Y^+ value of less than 4 or 5 is acceptable. Regarding the first layer thickness, the y value of 8×10^{-5} m is set in sHM tool. This is three times more than the first y value (2.5×10^{-5} m) and the aspect ratio is approximately 32. It is noteworthy that when the CFD simulation is analyzed in a low Reynolds turbulence model integrated the wall function to solve the near wall region (k-ω), the CFD simulation requires the first cell center needed to be placed at the viscous sublayer. That means the Y^+ value must be equal to 1 corresponding to the aspect ratio of 100. As a result, the mesh with the larger aspect ratio is chosen. Moreover, it can be assured that the Y^+ value equals 1 and the total layer thickness is 1.67 mm with 9 boundary layers and the expansion ratio of 1.3 (Table 2).

It is a fact that the more acute the geometry's angle, the more challenging it is to generate layers. While the layers along the tricopter's arms may not fully develop at the corners of the sectional rectangular shapes, the layers fully developed across the wing's chordwise direction demonstrate acceptable accuracy in this study, as displayed in Figures 8 and 9. Furthermore, the formation of layers around the fuselage are also acceptable (see Figures 5 and 7).



(a) (b)

Figure 8 Layers around the leading edge (a) and two arms (b)



(a) (b)

Figure 9 Layers around the horizontal stabilizer (a) and the fixed wing (b)

III. SIMULATION SETUPS

Initial conditions and boundary conditions are used in the numerical approach for solving the Navier-Stokes equation and continuity equation. This section introduces boundary conditions, turbulence modelling coefficients, and ways of specifying direction of flow and aerodynamic forces.

3.1 Boundary condition

The actual UAV is capable of cruising at a speed of 15 m/s. Therefore, the inflow speed at the inlet boundary is set to 15 m/s, while the out-flow boundary condition is used to model flow exiting at the outlet patch, with the pressure and velocity calculated based on the inlet boundary conditions (see Figure 10). The far-field boundaries including lower, upper, front, and back boundaries are set to the same velocity condition as the inlet boundary. To model the relative change in the incidence angle of attack, the flow directions are defined by specifying the speed components along the x-, y-, and z-axes, while the UAV location and the initial mesh generation remain unchanged. More details on defining the flow direction corresponding to various wing angles are presented in Section 0.

airiti

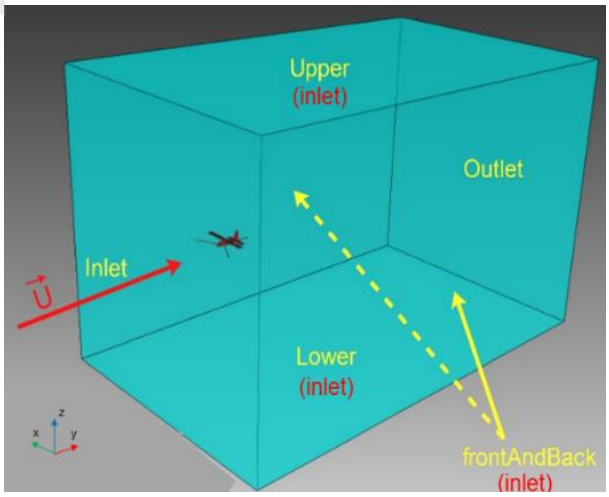


Figure 10 Boundary conditions

3.2 Turbulence model

To select the turbulence model, the Reynolds number is computed using the formula Re = $\rho V c/\mu$, where ρ is fluid density, V is cruising speed, c is the chord length, and μ is dynamic viscosity of the fluid. In this study, the Reynolds number is approximately 142,000 ~ 1.4×10^5 (computed based on parameters presented in Table 1). Consequently, the K-ω SST turbulence model is chosen to model the turbulent flow dissipation. In addition, by comparing the K-ω SST and K-ω transition SST turbulence models in our previous work [6], the current selection of K-ω SST model is validated. The findings indicated that K-ω SST model could provide results that are closer to the measured data when the Reynolds number is larger 7×10^4. In contrast, the K-ω transition SST model exhibited a 15% difference of C_D compared to the measured one at Re = 7×10^4, reported by Dong et al. [10]. In the K-ω SST model, the first transported variable is turbulent kinetic energy, k (m^2s^{-2}), and the second one is the specific dissipation, ω (ms^{-2}). These two parameters, as shown in Eq. (2), are specified as uniform values at the inlet, far-field, and wall boundaries, except for the outlet boundary, where a zero-gradient condition is applied. This indicates that the flow is fully developed in that region.

$$\begin{cases} \omega = 63.888\ ms^{-2} \\ k = 0.135\ m^2s^{-2} \end{cases} \tag{2}$$

3.3 Flow speed components

In this section, the relative wind velocity, lift, and drag directions are presented. The airspeed components in x-, y-, and z-axes are identified corresponding to different AoA. Figure 11 describes the UAV's positioning when pitching up or down to adjust the air speed. It is assumed that the wind is perpendicular to the Oxz plane, and the magnitude of inlet velocity, with an inlet velocity magnitude of 15 m/s. The flow speed components (U_x ,U_y, U_z) are defined in the controlDict file of OpenFOAM, corresponding to (0, U×cos(α), -U×sin(α)) m/s, where U is 15 m/s and α is the specific AoA.

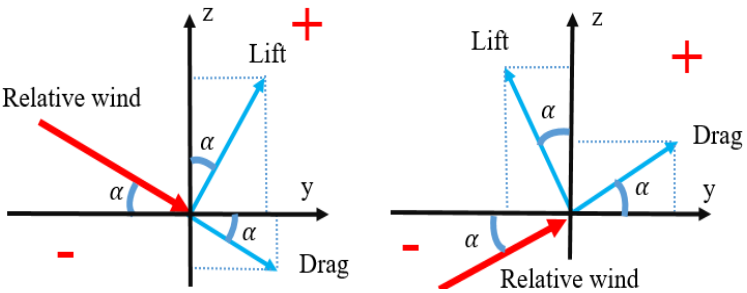


Figure 11 Flow vectors – positive AoA

IV. SIMULATION ANALYSIS RESULTS

4.1 Residual convergence

The low-Reynolds, steady-flow simulation is carried out using the SIMPLE solver and the K-ω SST transition turbulence model. These figures below illustrate the convergence histories at different AoA. Although the residuals for all field pressure (p), velocity (U), turbulence kinetic energy (k) and specific dissipation rate (ω) are set to 10^{-7}, most cases cannot meet the required residuals. Their residuals are in the order of 10^{-3} to 10^{-6}, showing that the results are acceptable. Additionally, the results reveal that the simulation requires more time to achieve stable

airiti

solutions, as the plots show significant fluctuations, particularly at larger AoA.

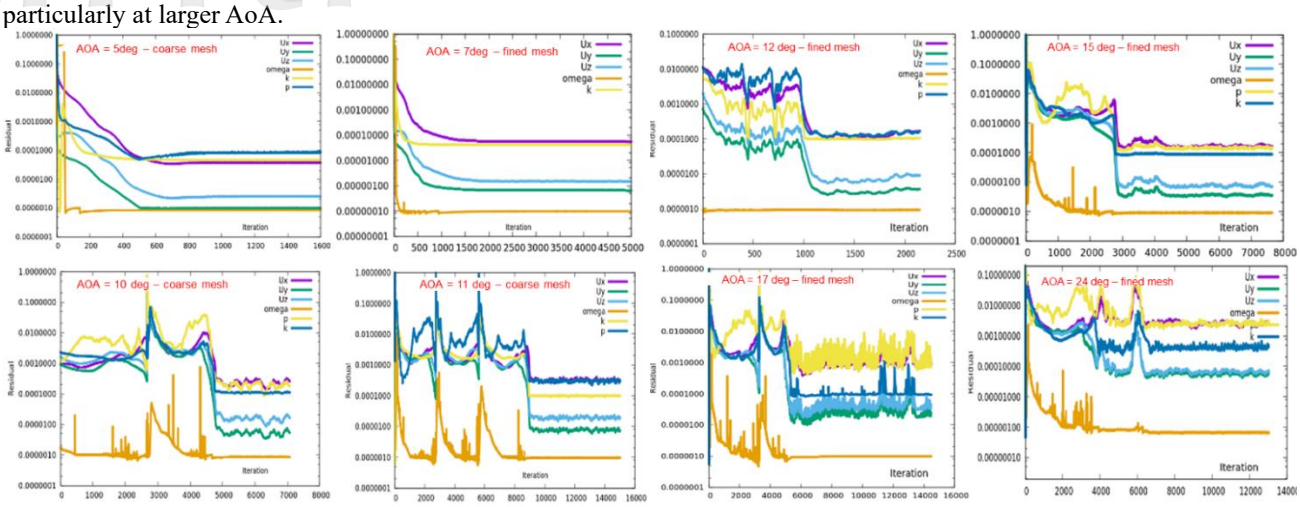


Figure 12 Residual curves for different AoA by the coarse and fined meshes.

4.2 Lift and drag coefficients

Figure 13 compares the results obtained by the analytical and numerical approaches using different software including Profili, OpenFOAM, and ANSYS-Fluent. Profili, a software based on the classical vortex lattice method, is used to calculate the lift and drag coefficients for the BE12355D airfoil, which is employed in constructing the fixed wing of the UAV-HOPE. The OpenFOAM and ANSYS-Fluent are also used to investigate the aerodynamic performance of the 2D airfoil and the 3D fixed wing. The 2D analysis of the BE12355D airfoil in a range of the low AoA from 0 to 7° reveals the similar results obtained from both Profili and OpenFOAM. Additionally, although the maximum lift coefficients show about 10% difference (comparing 1.42 and 1.55, see Figure 13), their maximum values are consistently confirmed at 10 o in both calculations. In the series of 3D simulations, both coarse and fined meshes illustrate a slight decrease in lift coefficient within the AoA range of 8 and 12°. However, the simulation performed by the coarse mesh cannot get the converged results, when the AoA is larger than 13°. The maximum lift coefficient of the 3D UAV with fixed wing, computed using the fine mesh, is approximately 1.39 at the AoA of 17°. In addition, the plot illustrates a steeper slope of the lift coefficient curve for the 2D airfoil in contrast to the shallower slope of the 3D wing. Furthermore, the larger stalling angle for the 3D fixed wing compared to that of the 2D airfoil, (17° versus 10°), confirms the numerical predictions for lift coefficients in both 2D and 3D analyses.

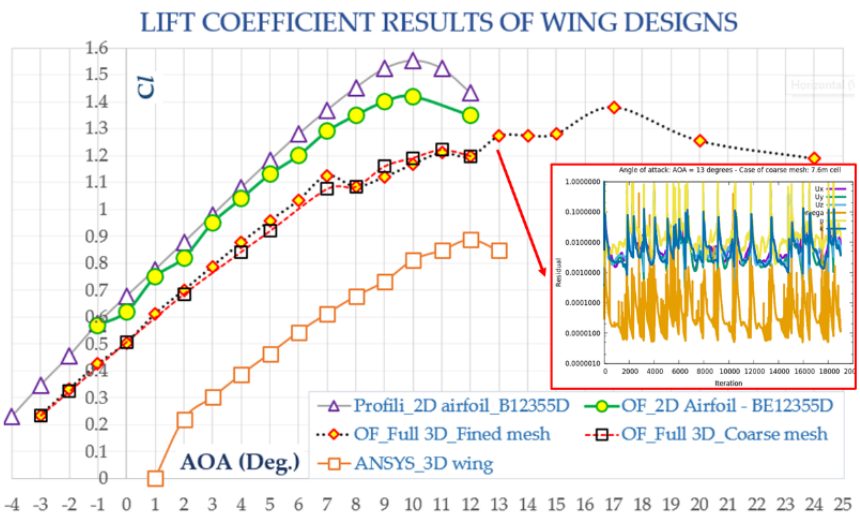


Figure 13 Comparison of lift coefficient results. (Note that "OF" stands for OpenFOAM)

In addition, the lift coefficient of 0.5039 is obtained at zero-degree AoA for the fined mesh 3D simulation. The fixed wing generates the lift as a part of total lift that is estimated as follows:

$$L = 0.5 \times \rho \times C_L \times V^2 \times S \quad (3)$$

$$\Rightarrow L = 12.5 \text{ N} \quad (4)$$

airiti

$$\Rightarrow \frac{L}{W} = \frac{12.5}{3.5 \times 9.81} = 36.40\% \quad (5)$$

Considering the wing lift-to-total lift ratio k_W of 0.4, as stated in Table 1, the computed lift-to-weight ratio of 0.36 indicates that the wing lift generation is predicted to be slightly lower (about 9%) than the designed value of 40% of the take-off weight, which corresponds to 3.5 kg.

$$\Rightarrow \text{Error} = \frac{|0.364-0.4|}{0.4} = 8.98\% \quad (6)$$

In the preliminary design phase, the lift coefficient is estimated, using the characteristics of the 2D airfoil. Meanwhile, the CFD simulation incorporates the 3D wingtip vortex phenomenon. Therefore, the CFD-based k_W should be lower than 0.4, indicating that the accuracy of the CFD simulations is acceptable.

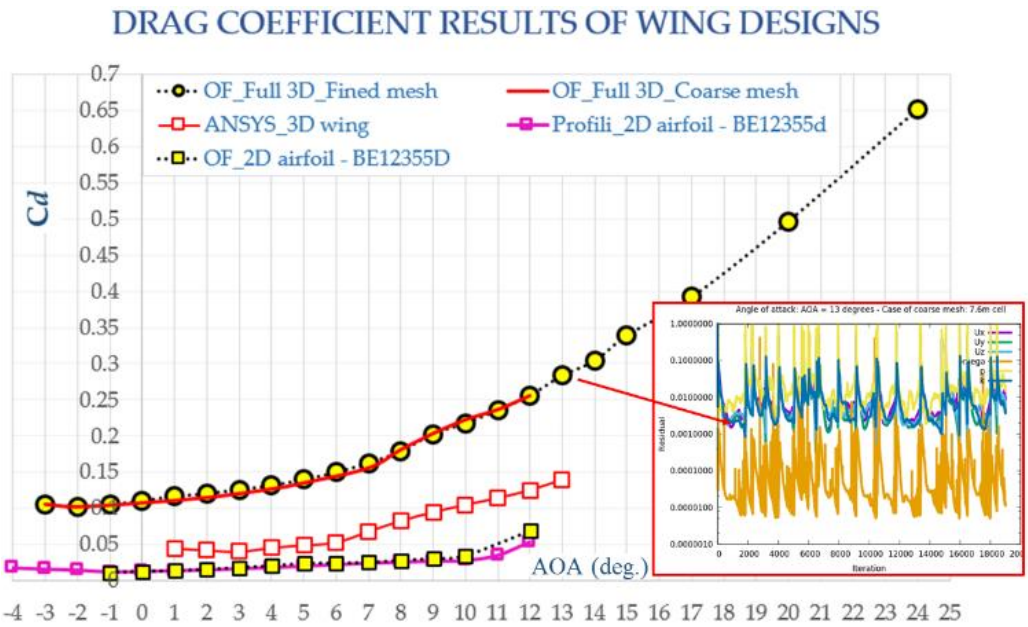


Figure 14 Comparison of drag coefficients by different analytical and simulation approaches. (Note that "OF" stands for OpenFOAM)

The drag increases once the UAV performs a pitching motion. Taking this into account, a necessary high AoA is essential for landing or altering its altitude. In more detail, Figure 14 describes the drag coefficients in relation to changes in AoA. It can be observed that that the UAV drag forces are not significantly dependent on either the grid size or the layer thickness within the low range of AoA from 0 to 10°. However, only the fined mesh can provide the drag coefficients at high AoA in an oscillating manner.

4.3 Overall flow fields

The flow fields for the UAV-HOPE are analyzed at moderate AoA of 4° where air flow attaches to the wing surfaces from leading edge to trailing edge and high AoA 12° where flow separation may occur.

4.3.1 Pressure distribution

In Figure 15, the fixed wing is investigated for pressure distribution in four spanwise sections (y/b): at 1/6, 1/2, 5/6 of wingspan and at the wing tip and three chordwise positions (x/c): 12%, 60% and 80% of the chord length. The pressure coefficients shown in Figure 16 and Figure 17 for AoA of 4° are the data extracted using paraView tool. The first group consists of sections, which are cut in the chordwise direction (Figure 16), where C_p is plotted from the leading edge to trailing edge for four spanwise locations. The negative values of C_p indicate that the air flow accelerates on the upper wing, with the acceleration decreasing from the wing root to the wingtip. In other words, along the spanwise direction, the relative flow speed increases from root to tip on the upper surface, while the pressure difference between the upper and lower wings also decreases. The second group comprises three chordwise sections, which are cut along the spanwise direction (Figure 17), where the pressure coefficient C_p is plotted across the entire wingspan. The flow quasi-symmetry can be observed from the pressure coefficients of the left and right wings. A high-pressure difference is recognized near the leading edge at the chordwise position of 12%c (represented by the green and red curves in Figure 17), gradually decreasing as the flow moves towards the trailing edge. Comparing the negative pressure coefficients on the upper wing surface at the three chordwise positions (x/c): 12%, 60% and 80%c reveals that the local flow speeds generally decelerate from the leading edge to trailing edge. However, the formation of the vortices near the wingtip leads to a significant reduction in static pressure. This results in the local velocity at the tip is higher at the trailing edge (represented by the yellow and blue curves at 80%c in Figure 17) compared to the upstream chordwise locations.

airiti

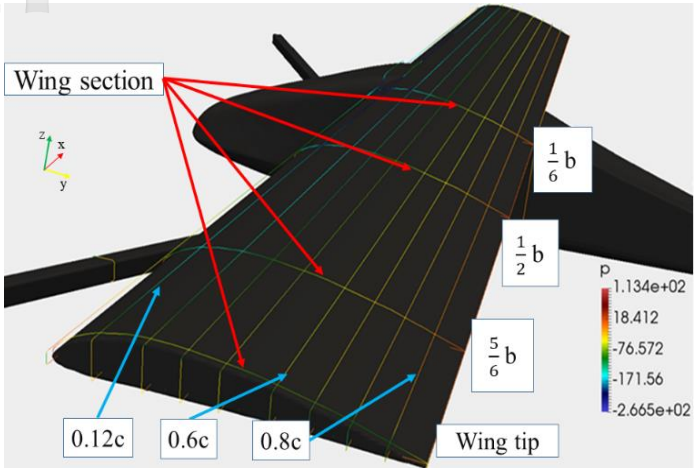


Figure 15 Cross sections considered for pressure distribution along the chord length and the wingspan

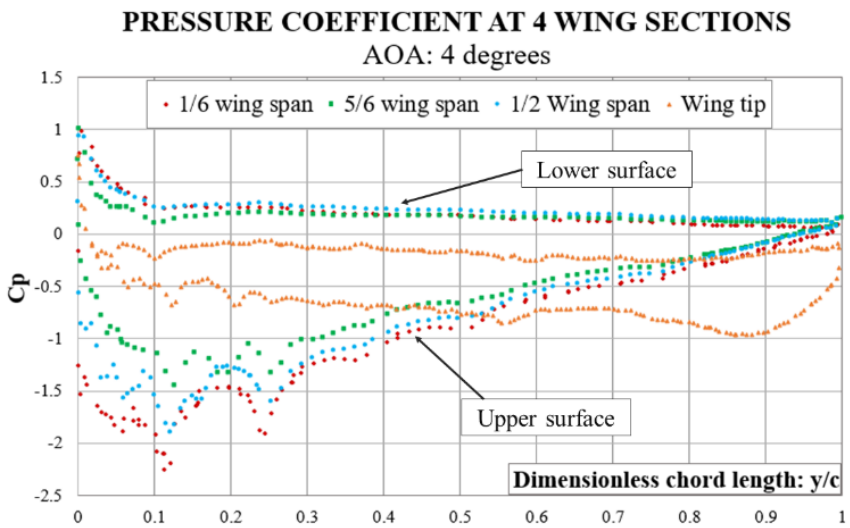


Figure 16 Pressure coefficient distribution along wingspan locations: 1/6, 1/2, and 5/6 of wingspan and wingtip (α = 4°).

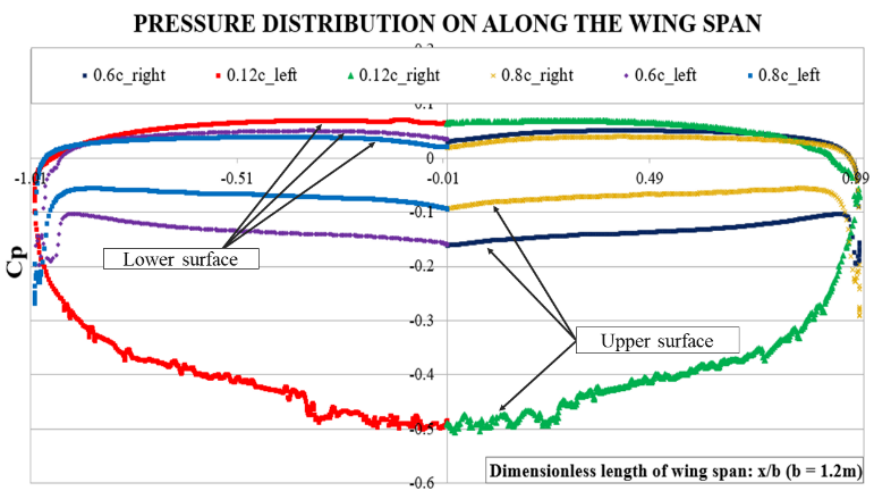


Figure 17 Distribution of spanwise pressure coefficients at chordwise positions from leading to trailing edge: 12%, 60%, and 80% of chord length

airiti

4.3.2 Velocity Field

At larger AoA of 12°, the flow is completely turbulent with the maximum speed of about 26 m/s along the curve of the anterior body of the UAV. Meanwhile, that is about 21 m/s when AoA is 4° (see velocity magnitudes shown by the color bars in Figure 18 and Figure 19a), while the pressure distribution on the surface is shown in Figure 19b. At x/b = 1/3 (see Figure 20), for both cases at AoA of 4° and 12°, the airflow velocity increases at the leading edge and reduces as it flows over the airfoil curvature. At 12°, the airflow tends to detach at approximately 50% of the chord length, as indicated by the color change from red to light yellow. Besides, the velocity of the airflow elements at the trailing edge of the wing is nearly zero and this region is wider when compared to that of the case at 4°. This indicates the flow become turbulent at the trailing edge when AoA is high.

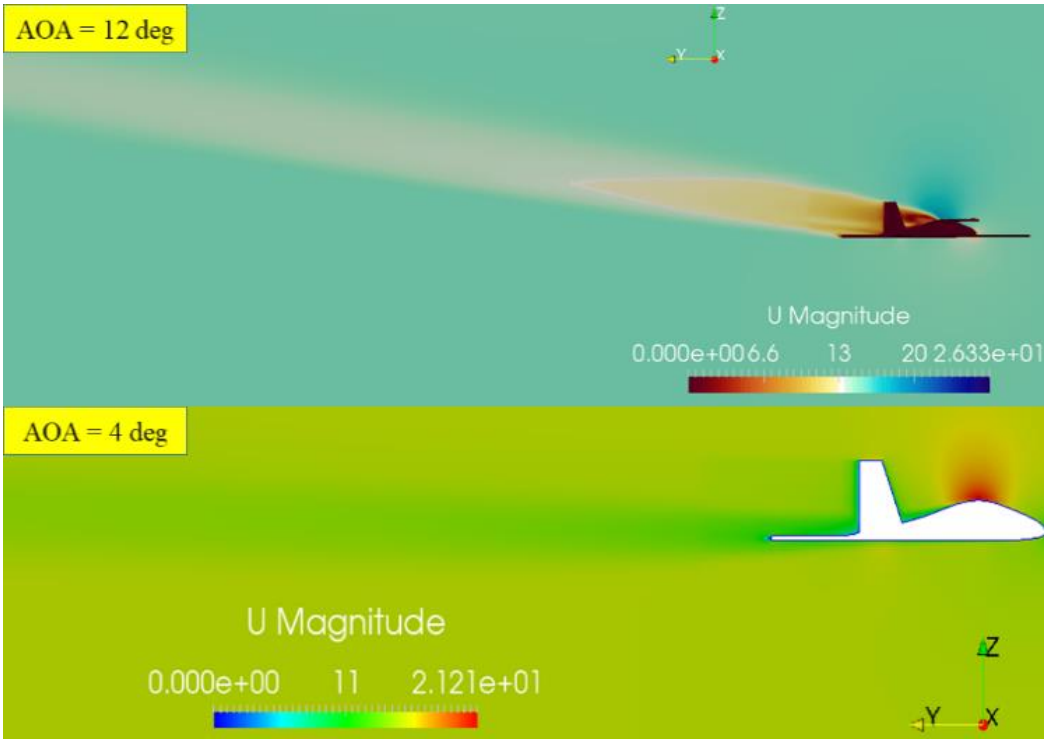
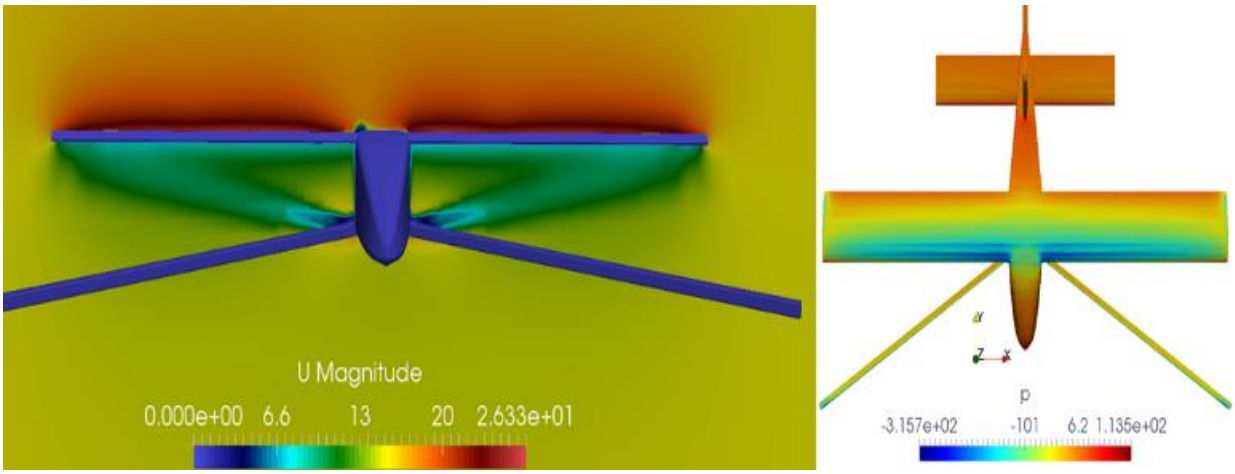


Figure 18 Velocity fields



(a) (b)

Figure 19 (a) Velocity field (α = 12°); (b) Pressure distribution on the UAV' surface (α = 4°)

airiti

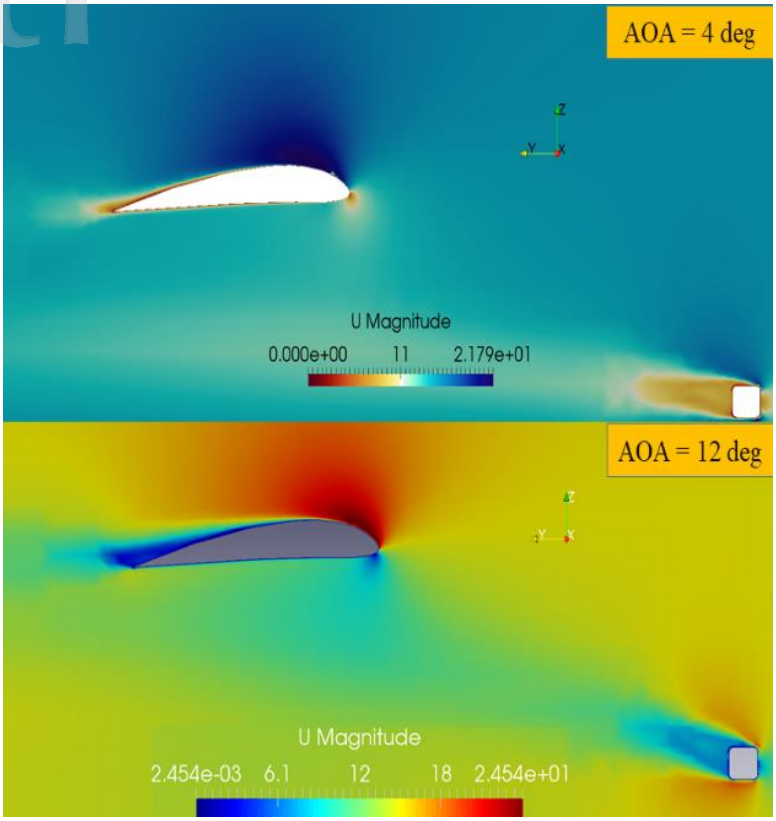


Figure 20 Velocity fields of the wing section at one-third of the wingspan (x/b =1/3)

4.4 Analysis of the laminar separation bubbles of the low Reynolds number flow

In this section, the airflow distribution around the upper and lower fixed-wing surfaces is presented for various spanwise positions. The flow transition phenomena and the laminar separation bubbles (LSB) on the wing's surface of a full 3D UAV are described in detail. LSB is a typical phenomenon of the low Reynolds number flow, particularly in near-space, low-speed aircraft and micro-air vehicles. This phenomenon may arise when the airflow encounters strong adverse pressure, causing the laminar flow to separate from the wing surface.

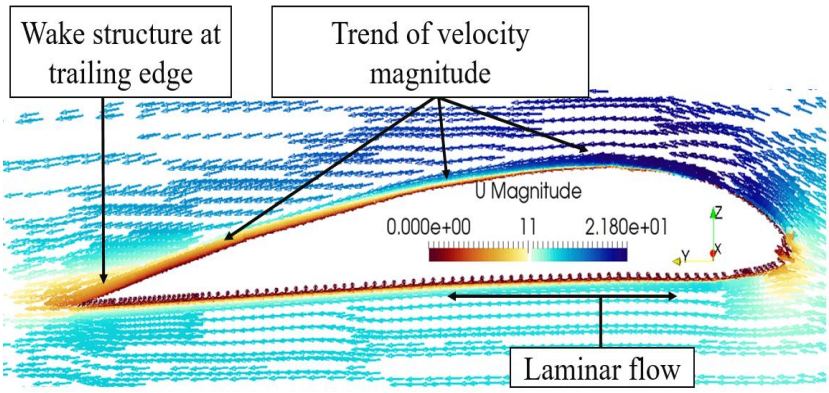


Figure 21 Air flows over the wing section (α = 4°) at 75% of the wingspan (x/b =3/4)

In the following discussions, the flow behaviors at the AoA of 4° and 12° are characterized by the streamlines across several wing sections, with the velocity magnitude represented according to the color bar. For example, Figure 21 illustrates the streamlines along the chord line at a wing section, which is located 75% of the wingspan, with the AoA of 4°. As depicted, the flow closely follows the upper contour of the airfoil from the leading edge to the trailing edge. This indicates that the flow remains attached to the wing with no occurrence of the LSB when the AoA is small. In Figures 22-27, the distinct structures of the LSB observed at the AoA of 12° are graphically demonstrated at various sectional wing locations of 1/6, 1/3, 1/2, 2/3, and 5/6 of the wingspan.

airiti

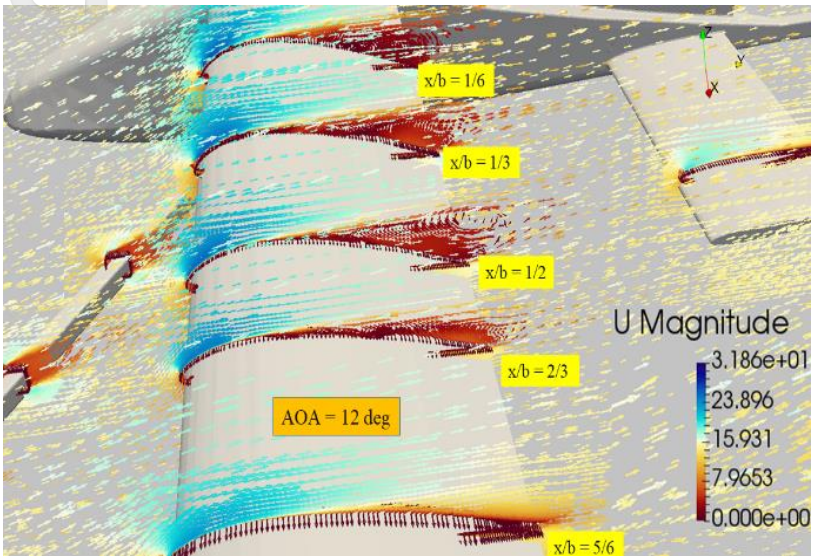


Figure 22 Flow characteristics distributed at 5 wing sections (α = 12°)

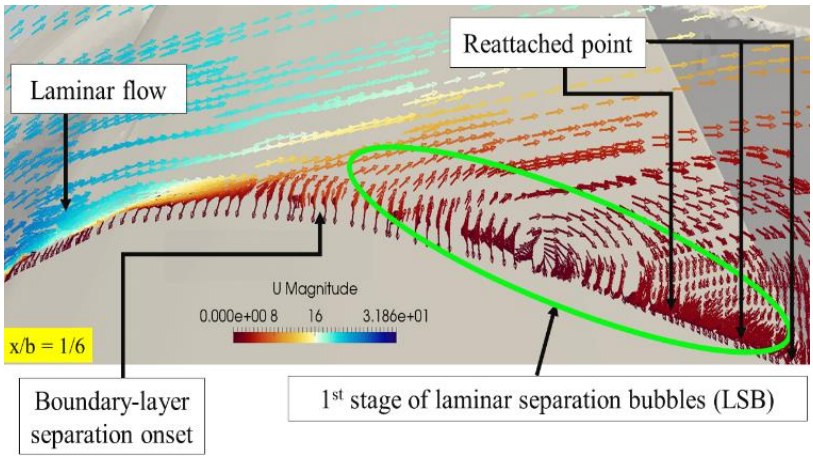


Figure 23 Airflow over wing surface exhibiting transition from laminar to turbulent situation at x/b = 1/6 (α = 12°)

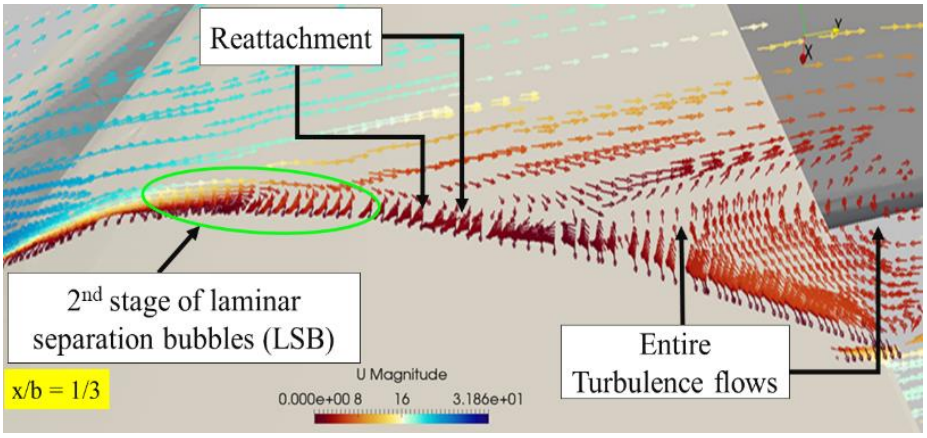


Figure 24 Flow behaviors at wing section at x/b = 1/3 (α = 12°)

airiti

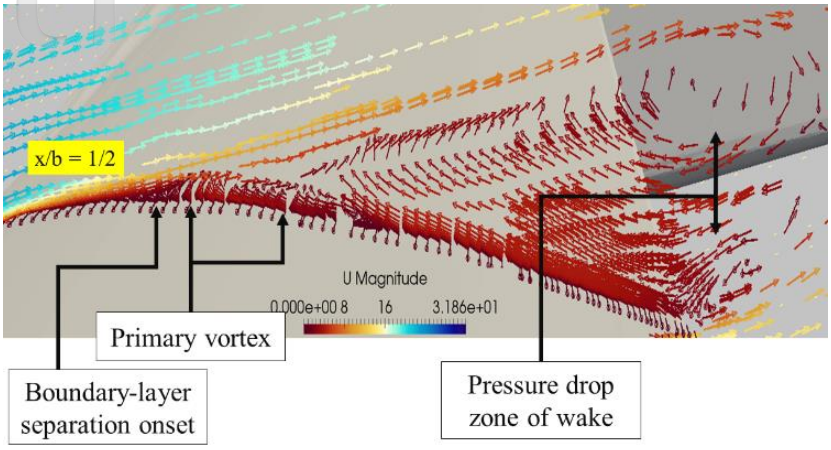


Figure 25 The entire turbulent flow at wing section x/b = 1/2 (α = 12°)

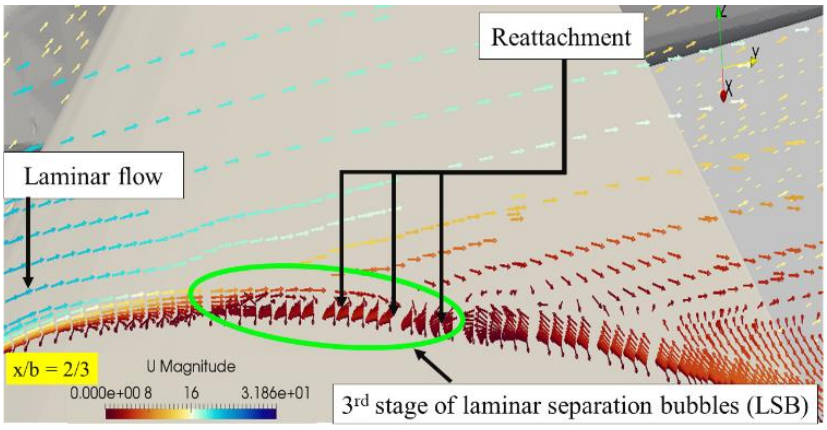


Figure 26 Flow at wing section x/b = 2/3 (α = 12°)

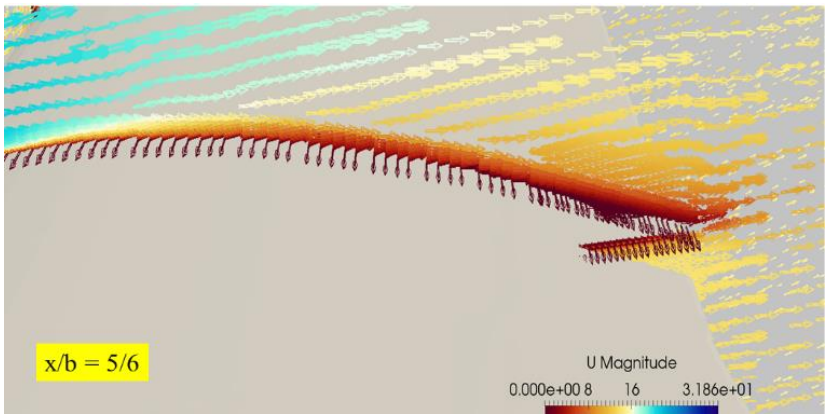


Figure 27 Wake structure at the trailing edge at x/b = 5/6 (α = 12°)

Specifically, at one-sixth of the wingspan shown in Figure 23, the flow separation occurs at 50% of the chord length, with a long, tall LSB depicted near the trailing edge. This wing section also features the largest size of the separation bubbles. At one-third of the wingspan in Figure 24, the flow separation is observed at approximately 70% of the chord length, while a long LSB extends earlier from 20% to 60% of the chord. As shown in Figure 25, the flow at 50% of the wingspan is completely separated at around 30% of the chord length, with no reattachment to the wing after separation. This shows that the LSB are not present in this section.

Furthermore, at 2/3 of the wingspan, the size of LSB decreases in both height and length, and flow separation

airiti

occurs at nearly 50% of the wing chord length after the flow lightly reattaches to the wing. Finally, due the effect of the wingtips, the flow around the wingtip, enhanced by high-pressure air beneath the wing surface flowing around the tip to the low-pressure area above, exhibits the large structures of vorticities (Figure 28). This explains why the LSB is not present at the wing tip, where flow separation does not occur (see Figure 27).

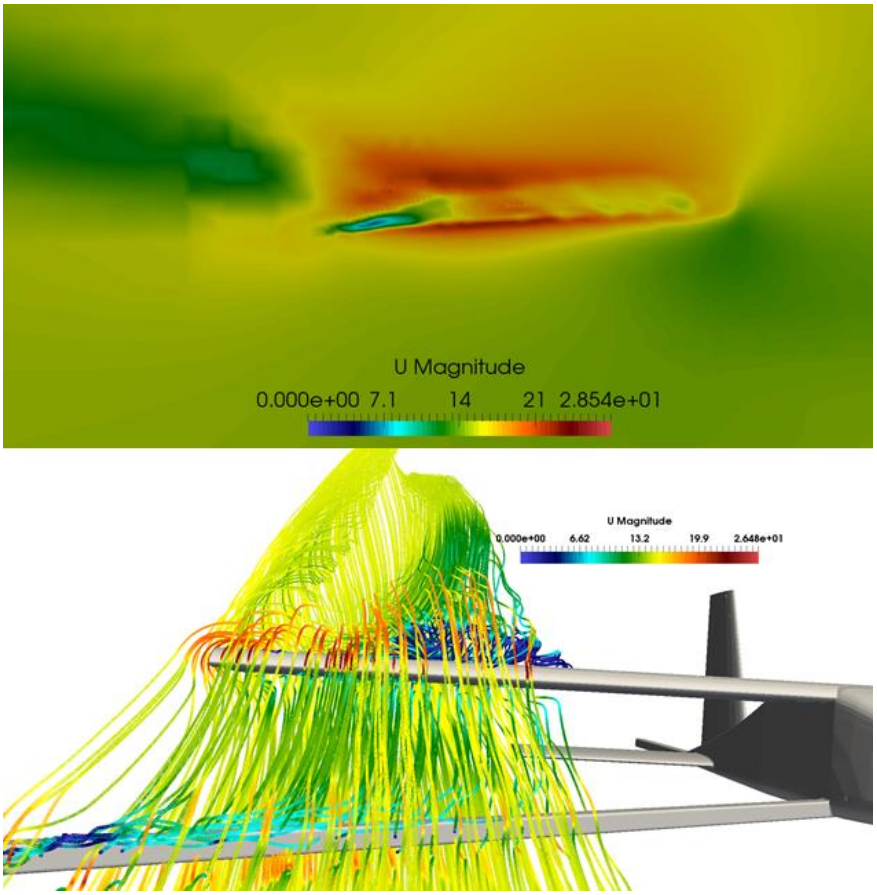


Figure 28 Wing-tip effect over the fixed wing at α = 12°

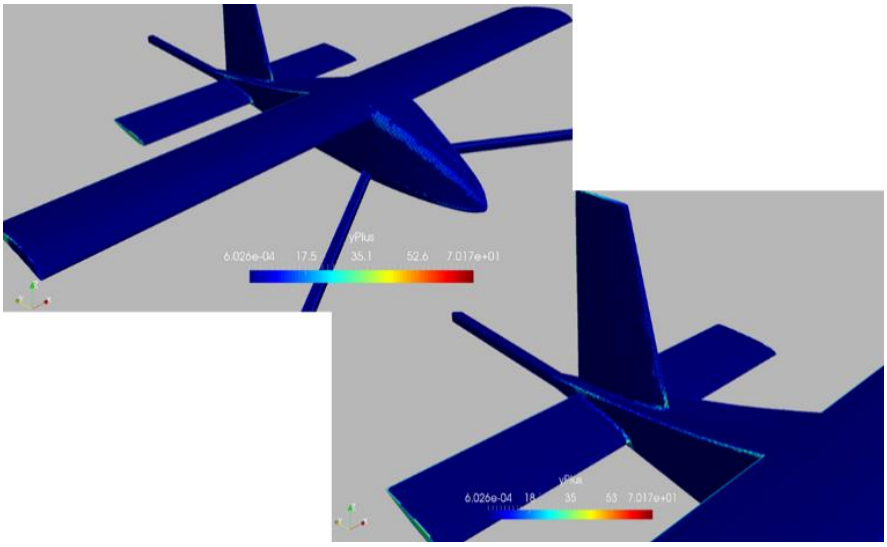


Figure 29 Y^+ field on the UAV's surface

Our observations align qualitatively with the findings presented in [9] for a finite wing, NACA 0018 airfoil of Reynolds number 1.25×10^6 and AoA of 6°. Deploying particle image velocimetry, the experimental results

airiti

highlight both the formation and suppression of the LSB extending from the inboard section of the wing to the tip, as well as local flow behaviors associated with these phenomena. This is notable that at a high AoA of 12°, the presence of the LSB intermittently along the wingspan helps delay the stall of the UAV wing and increases lift coefficient of the UAV model compared to the performance of the 2D airfoil at the same AoA.

4.5 Y plus fields

After finishing the simulation, the Y^+ values distributed on the UAV surface are demonstrated in ParaView tool.

Figure 29 shows the low Y^+ values colored by the dark blue which corresponds to about 1. Although some positions, such as the nose, two arms, and vertical stabilizers, have the Y^+ values greater than 10, the average Y^+ of the entire UAV is only 1.16. This ensures the reliability of the simulation results.

V. CONCLUSION

This paper successfully introduces a numerical method, which can reasonably predict the aerodynamic performance of a full UAV model. By predicting aerodynamic coefficients, the study could provide an insight into developing a novel design of a hybrid UAV model. Conventional steps for such a CFD method are carefully presented and discussed. By comparing the CFD results of 2D airfoil from different software, the findings reveal agreement in the stalling angles, as well as acceptable results for the maximum lift coefficients with a difference of within 10%. In the series of 3D simulations, the lift coefficient versus AoA curve indicates a stall occurred at a high AoA of 17° with a maximum lift coefficient of 1.4. Overall, the fixed wing cruising at a speed of 15 m/s can produce up to 40% of the lift required to compensate for the total weight of the UAV. The simulation indicates that the stall phase is appropriate for the 3D UAV model, and a slight adjustment of the AoA is necessary to achieve the required lift for varying payloads. In addition, due to production of induced drag associated with the finite wing and the parasite drag, the drag coefficient of the full 3D UAV model is larger than that of the BE12355D airfoil, which typically experience only profile drag. Moreover, on the wing upper surface, the laminar separation bubbles, which are typical phenomena of the low Reynolds number flow, can be recognized and analyzed. Despite lacking experimental validation, this study can be regarded as a feasible preliminary effort necessary for further analyses on the stability and controllability of the full 3D model, when the dynamic mesh technique is employed.

ACKNOWLEDGMENTS

We acknowledge Ho Chi Minh City University of Technology (HCMUT), VNU-HCM for supporting this study.

REFERENCES

[1] Kody F, Bramesfeld G, "Small UAV design using an integrated design tool," *International Journal of Micro Air Vehicles*, Vol. 4, No. 2, 2012, pp. 151-163. https://doi.org/10.1260/1756-8293.4.2.151

[2] Donadon MV, Frascino S, Iannucci L, "A vortex lattice program for steady state aerodynamic analysis of flapped and twisted uav wing plan forms," Brazilian Symposium on Aerospace Engineering & Applications, 2009. https://api.semanticscholar.org/CorpusID:27415692

[3] Kroll N, Rossow CC, Schwamborn D, Becker K, Heller G, "MEGAFLOW—a numerical flow simulation tool for transport aircraft design," 23rd Congress of International Council of the Aeronautical Sciences, 2002, Paper ICAS 2002-1.10.5 https://www.icas.org/icas_archive/ICAS2002/PAPERS/1105.PDF

[4] Sun CN, Yeh YL, "On the Experimental and Application Study of High-Curvature Wing for Load-Task UAV", *Journal of Aeronautics, Astronautics and Aviation*, Vol. 51, No. 4, 2019, pp. 369-379. https://doi.org/10.6125/JoAAA.201912_51(4).03

[5] Johari MS, Ali ZM, Wisnoe W, Ismail N, Ishak IS, "Computational Aerodynamic Analysis of UiTM's Hawkeye UAV Aircraft", *Journal of Aeronautics, Astronautics and Aviation*, Vol. 53, No. 2, 2021, pp. 295-302. https://doi.org/10.6125/JoAAA.202106_53(2).23

[6] Le THH, Huynh TP, Nguyen DA, Nguyen, NH, "Simulation of Drag Crisis Phenomena Over a 3D Teardrop-shaped Body," *Journal of Aeronautics, Astronautics and Aviation*, Vol. 55, No. 2, 2023, pp. 105-117. https://doi.org/10.6125/JoAAA.202306_55(2).01

[7] Schmitz FW, "Aerodynamics of the model airplane Part I, airfoil measurements, translated from German," Redstone Scientific Information Center RSIC-721, NASA Technical Report, 1967. https://ntrs.nasa.gov/citations/19700029685

[8] Xia T, Dong H, Yang L, Liu S, Jin Z, "Investigation on flow structure and aerodynamic characteristics over an airfoil at low Reynolds number - A review", *AIP Advances*, Vol. 11, 050701, 2021. https://doi.org/10.1063/5.0044717

[9] Toppings CE, Yarusevych, S, "Structure and dynamics of a laminar separation bubble near a wingtip," *Journal of Fluid Mechanics*, Vol. 929, 2021, p. A39. https://doi.org/10.1017/jfm.2021.881

[10] Dong P, Miau JJ, Zoghlami A, "An experimental study about drag crisis phenomenon on teardrop model", *Journal of Aeronautics, Astronautics and Aviation*, Vol. 51, No. 2, 2019, pp. 141-157. https://doi.org/10.6125/JoAAA.201906_51(2).01

airiti

[11] Pham HT, Le THH, "Stability analysis and simulation for vertical take-off and, landing drones in hover," Student Scientific Research Conference, Ho Chi Minh City University of Technology, Vietnam, 2018.

[12] Tran DKK, Nguyen K, Le THH, Nguyen NH, "Numerical Simulation for The Forward Flight of the Tri-copter Using Virtual Blade Model," *Journal of Advanced Research in Fluid Mechanics and Thermal Sciences*, Vol. 67, Issue 1, 2020, pp. 1-32. https://doi.org/10.37934/arfmts.67.1.132

[13] Le MQ, "Numerical Simulation of the Hybrid UAV of tri-copter and fixed-wing (HOPE)," Graduation Thesis, Ho Chi Minh City University of Technology, Vietnam, 2018.

[14] Tran TXH, Numerical Simulation of the Hybrid UAV of tri-copter and fixed-wing (HOPE)," Graduation Thesis, Ho Chi Minh City University of Technology, Vietnam, 2018.

[15] ESI-OpenCFD, "4.4 Mesh generation with the snappyHexMesh utility," Accessed 29 September 2024 [Online] www.openfoam.com/documentation/user-guide/4-mesh-generation-and-conversion/4.4-mesh-generation-with-the-snappyhexmesh-utility.

[16] Hu YF, Carter JG, Blake RJ, "The effect of the grid aspect ratio on the convergence of parallel CFD algorithm", Parallel Computational Fluid Dynamics 1995, 1996, pp. 289-29. https://doi.org/10.1016/B978-044482322-9/50090-6